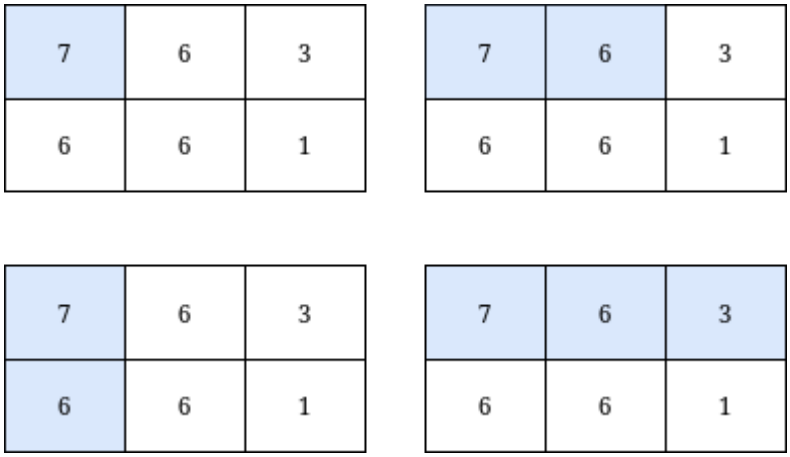
3070. Count Submatrices with Top-Left Element and Sum Less Than k

You are given a **0-indexed** integer matrix grid and an integer k.

Return *the **number** of submatrices that contain the top-left element of the* grid, *and have a sum less than or equal to* k.

Example 1:

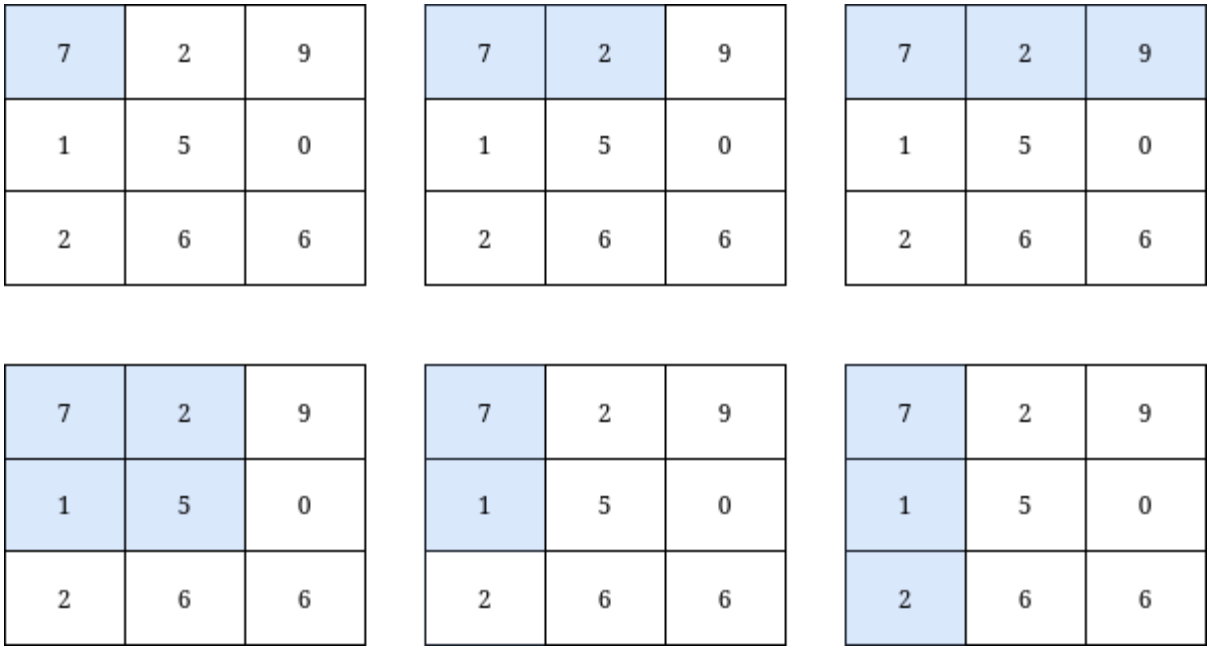


Input: grid = [[7,6,3],[6,6,1]], k = 18

Output: 4

Explanation: There are only 4 submatrices, shown in the image above, that contain the top-left element of grid, and have a sum less than or equal to 18.

Example 2:



Input: grid = [[7,2,9],[1,5,0],[2,6,6]], k = 20

Output: 6

Explanation: There are only 6 submatrices, shown in the image above, that contain the top-left element of grid, and have a sum less than or equal to 20.

Constraints:

- m == grid.length
- n == grid[i].length
- 1 <= n, m <= 1000
- 0 <= grid[i][j] <= 1000
- 1 <= k <= 10^9

Solution:--

```
class Solution {

    public int countSubmatrices(int[][] grid, int k) {
        int n = grid.length;
        int m = grid[0].length;
```

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        int[] cols = new int[m];
        int res = 0;

        for (int i = 0; i < n; i++) {
            int rows = 0;
            for (int j = 0; j < m; j++) {
                cols[j] += grid[i][j];
                rows += cols[j];
                if (rows <= k) {
                    res++;
                }
            }
        }

        return res;
    }
}
```